APOORVA BUSUNUR MALLIKARJUNA



MSc in Aerospace Engineering (Embedded Systems), ISAE-SUPAERO, France

apoorvabm6.github.io | apoorva-bm | ApoorvaBM6 | Toulouse, France
bmapoorva.nr@gmail.com | +33 7 45 37 97 17 | +91 96 11 58 97 99

EXPERIENCE

Research Intern

Neuroergonomics Lab, ISAE-SUPAERO

Apr 2025 – Sep 2025 | Toulouse, France | Report

Developed a **real-time blink detection algorithm** from 2-channel fronto-polar EEG signals to monitor mental fatigue in professionals operating in extreme environments.

- Designed and conducted human-subject experiments.
- Implemented the end-to-end EEG acquisition and processing pipeline.
- **Skills:** Research Methodology, Python, EEG Signal Processing, Machine Learning.

Software/Firmware Engineer I & II

Hewlett Packard

Oct 2020 – Aug 2023 | Bangalore, India | Recognitions

Developed **Control Panel UI features** for 40+ enterprise printer products with responsibilities spanning requirement analysis, feature implementation, and system integration.

- Acted as point-of-contact for UI team during feature rollouts and product line deployment(Coaster, performing system triage, debugging, and customer escalation handling.
- Promoted to Engineer II in June 2023.
- **Skills:** JavaScript, C#, UI Development, System Debugging, Requirement Analysis.

PROJECTS

Research Project: Low Cognitive Load Interface Design

ISAE-SUPAERO

Dec 2025 – Feb 2026 | Toulouse, France | Report

A Survey of Interaction Techniques for **Low Cognitive-Load Based Swarm Teleoperation**

- Conducted a structured literature review focusing on during 2015-2025.
- Compared interaction modalities, workload metrics, and evaluation approaches across the literature.
- Synthesized the findings into evidence-based design recommendations for interfaces that reduce operator cognitive load during swarm supervision.

Autonomous Drone

ISAE-SUPAERO

Nov 2024 – Feb 2025 | Toulouse, France | Report

Implemented a **skill-based architecture** with a planning and decision-making layer in ROS 2 for autonomous systems.

- Integrated object detection, autonomous navigation, and mission planning.
- Awarded the maximum grade for proactive analysis and detailed report.
- **Skills:** ROS2, AI Planning, Gazebo, Autonomous Systems.

Vision-Based Autonomous Drone (IMAV 2024)

IONLab, ISAE-SUPAERO

Jan 2024 – Dec 2024 | Toulouse, France | Presentation

Developed a quadrotor for vision-based **autonomous navigation in GNSS-denied environments** for IMAV 2024.

- Computer vision-based path detection(Canny Edge Detection and Hough Line Transform)
- Real-time autopilot communication through MAVLink.
- **Skills:** NVDIA Jetson Nano, OpenCV, PX4, C++, Python, CMake.

EDUCATION

M.Sc in Aerospace Engineering (Major : Embedded Systems)

ISAE-SUPAERO, Toulouse, France

Sept 2023 – Feb 2026 | Course List

B.E. in Electronics and Communication Engineering

The National Institute of Engineering, Mysuru, India

Aug 2016 – June 2020 | Course List

SKILLS

Programming & Methods

Python | C++ | JavaScript | EEG Signal Processing | Machine Learning | Computer Vision

Tools & OS

Visual Studio | Microsoft Office | GitHub | MATLAB | CMake | JIRA | LaTeX | Zotero | Linux | Windows | RTOS

Soft Skills

Scientific Curiosity | Collaboration | Analytical Thinking | Autonomy | Problem solving | Perseverance

LANGUAGES

English
French
Hindi
Kannada

ACTIVITIES

- **BR41N.IO BCI Hackathon 2026** – Participant
- **ICCAS 2026** – Volunteer & Participant
- **CIME 2025** – Volunteer & Participant
- **IMAV 2024** – Paper Presenter
- **ISAE-SUPAERO Embedded Systems Cohort** – Student Representative, 2024–2025

INTERESTS

Human Spaceflight | Neuroergonomics | Human-Centered Innovation | Swimming | Travel | Cooking